

Table-Gen RAG

A Retrieval-Augmented Generation Framework for
Automated Clinical Trial Table Shells

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The Problem

Why Clinical Reporting is a Multi-Month Bottleneck

| The Current Manual Bottleneck

Manual Table Authoring

- ✓ Statisticians produce **100+ table shells** per study.
- ✓ Derived manually from SAP, CRF, and CSP documents.
- ✓ Highly repetitive and error-prone process.
- ✓ Typically requires **several months** of dedicated work.



| Redefining Timeline Efficiency

Months

Traditional Manual Process



Days

Table-Gen RAG Pipeline

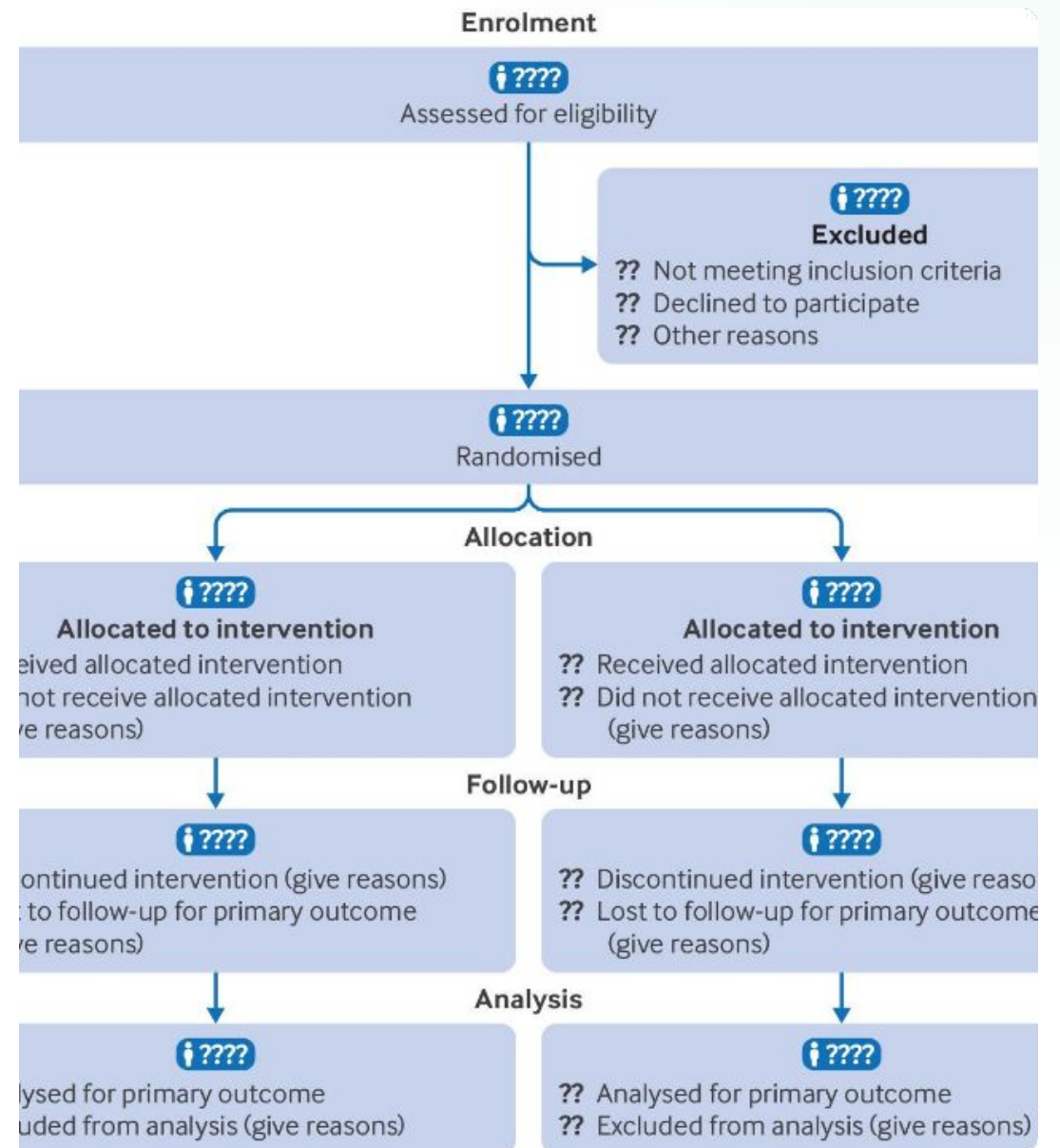
"Our solution streamlines clinical table shell generation by automating variable extraction and mapping."

Table-Gen RAG Framework

Unified Dual-Component Architecture

Utilizing **Llama-3.3 70B** served via Groq API for high-speed, high-precision retrieval and generation.

- ✓ **Variable Extractor:** Ingests SAP & CRF documents.
- ✓ **Table Generator:** Intelligent mapping to company templates.
- ✓ **Human-in-the-Loop:** Real-time editing interface.



Component 1: Variable Extractor



SAP Ingestion

Reads Statistical Analysis Plans to identify primary and secondary endpoints.



CRF Analysis

Extracts data collection variables directly from Case Report Forms.



Category Alignment

Systematically organizes variables into hierarchical trial categories.

Component 2: Table Shell Generator

Intelligent Template Mapping

- ✓ Identifies target shells from standardized corporate templates.
- ✓ Populates table headers and stubs with extracted variables.
- ✓ Outputs structured tables with precise placeholder values.
- ✓ Maintains compliance with CDISC/company standards.



The Interactive Advantage



Human-in-the-Loop Validation

Unlike "black box" generators, Table-Gen RAG provides an editable web interface for statisticians.

- ✓ **Real-time Refinement:** Add, remove, or reorder variables dynamically.
- ✓ **Validation:** Humans verify the LLM's mapping before final export.
- ✓ **Efficiency:** No need to restart the pipeline for minor adjustments.

Engineering Challenges & Solutions

Technical Challenge	Our Engineering Solution
Hierarchical PDF Parsing	Transitioned from 'unstructured' to CSV pre-processing for perfect template fidelity.
Complex Table Stubs	Utilized Pandas-based table readers to maintain multi-level row integrity.
Retrieval Precision	Guided Llama-3.3 to contextually cross-reference SAP categories with template stubs.

The Future Roadmap

Hierarchical-Hybrid RAG

Section-level retrieval combined with fine-grained search.

Component 3: Auto-Data

Automated population of shells with raw clinical datasets.

Semantic Terminology

Integrated keyword matching for clinical abbreviations.

Full End-to-End

Zero manual conversion from PDF to final report.

Questions?

Thank you for your attention.

 github.com/Sravya-Pogiri/table_shells
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